Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. **Sample the data and combine the files**

Took a small percentage of entries for pickup in every hour of a date. So, for all the days in a month, iterated through the hours and selected 5% values randomly from those. Used `tpep_pickup_datetime` for this.

1.1.2. Saved the sampled data to parquet file on drive to be used for further analysis

1.1.3. Loaded the trip_df dataframe from the data saved from all hours and days

```
trip_df.info()
```
✓ 0.0s

```
<class 'pandas.DataFrame'>
RangeIndex: 1896400 entries, 0 to 1896399
Data columns (total 20 columns):
 #   Column                 Dtype
---  ------                 -----
 0   VendorID               int64
 1   tpep_pickup_datetime   datetime64[us]
 2   tpep_dropoff_datetime  datetime64[us]
 3   passenger_count        float64
 4   trip_distance          float64
 5   RatecodeID             float64
 6   store_and_fwd_flag     str
 7   PULocationID           int64
 8   DOLocationID           int64
 9   payment_type           int64
 10  fare_amount            float64
 11  extra                  float64
 12  mta_tax                float64
 13  tip_amount             float64
 14  tolls_amount           float64
 15  improvement_surcharge  float64
 16  total_amount           float64
 17  congestion_surcharge   float64
 18  airport_fee            float64
 19  Airport_fee            float64
dtypes: datetime64[us](2), float64(13), int64(4), str(1)
memory usage: 291.3 MB
```

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

Striped the column names. Reset index of trip_df dataframe. Dropped column strore_and_fwd_flag

2.1.2. Combine the two airport_fee columns

Filled missing values in airport_fee with the values in Airport_fee column. Dropped the redundant column Airport_fee

2.1.3. Fixing negative values columns

Found the columns with negative values. Observed the total_amount column is sum of rest of fare related numeric columns. Replaced negative values to absolute values for the columns - total_amount', 'improvement_surcharge', 'mta_tax', 'congestion_surcharge', 'airport_fee', 'extra'

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column:

Below 5 columns have missing values (given with proportion)

store_and_fwd_flag 0.034209
RatecodeID 0.034209
passenger_count 0.034209
airport_fee 0.034209
congestion_surcharge 0.034209

2.2.2. Handling missing values in passenger_count

Number of records missing passanger count is 64874. Imputed the missing values for **passenger_count with it's median**

2.2.3. Handle missing values in RatecodeID

Number of records missing values in RatecodeID – 64874. Imputed the RetecodeID missing values with it's mode

2.2.4. Impute NaN in congestion_surcharge

Missing values for congestion_surcharge 64874. Imputed the missing congestion_surcharge values with it's mean.

2.2.5. Replaced 'airport_fee' missing values with it's mean

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

Potential numeric outliers (IQR method):

trip_distance: 249302 potential outliers (below -2.47 or above 6.92)

fare_amount: 197413 potential outliers (below -9.60 or above 40.80)

tip_amount: 145673 potential outliers (below -4.13 or above 9.55)

total_amount: 218083 potential outliers (below -6.51 or above 53.41)

trip_duration: 113414 potential outliers (below -11.86 or above 40.21)

extra: 34047 potential outliers (below -3.75 or above 6.25)

tolls_amount: 153802 potential outliers (below 0.00 or above 0.00)

improvement_surcharge: 2181 potential outliers (below 1.00 or above 1.00)

congestion_surcharge: 205772 potential outliers (below 2.50 or above 2.50)

airport_fee: 225836 potential outliers (below 0.00 or above 0.00)

Handling outliers:

Total records before handling outliers - 1896400

1. Removed records with passanger count greater than 6, as there are only 21such records which may skew stats for the column
2. Removed records for trip_distance == 0 , fare_amount == 0 and different zones – 63 such records – looks invalid data
3. Removed records for fare_amount > 300 and trip_distance < 0.5 – 36 records – looks invalid data
4. Keep only trips with pickup year 2023 – 1 record removed

Total records after handling outliers - 1896233

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

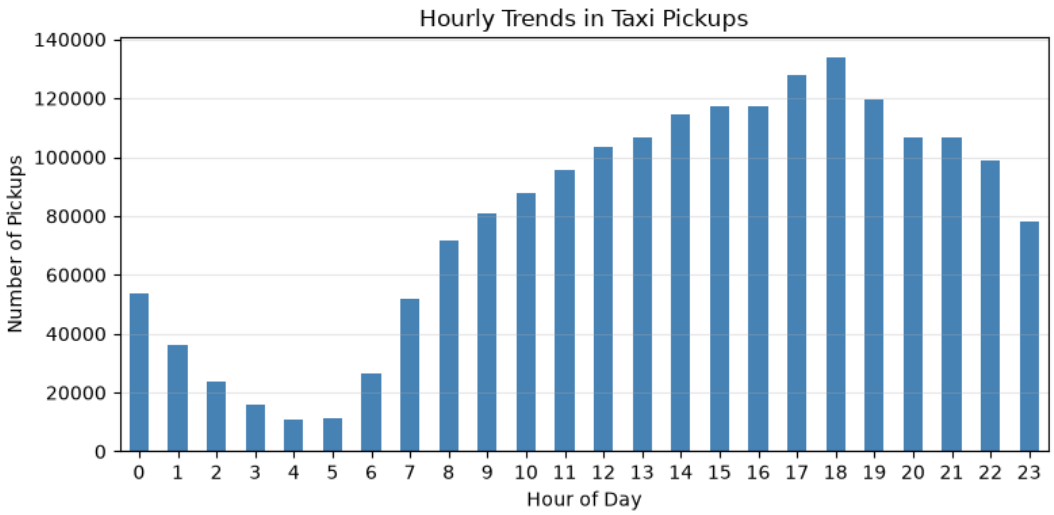
Numerical variables: ['passenger_count', 'trip_distance', 'trip_duration', 'fare_amount', 'extra', 'mta_tax', 'tip_amount', 'tolls_amount', 'improvement_surcharge', 'total_amount', 'congestion_surcharge', 'airport_fee']

Categorical variables: ['VendorID', 'RatecodeID', 'PULocationID', 'DOLocationID', 'payment_type', 'store_and_fwd_flag']

Temporal variables: ['tpep_pickup_datetime', 'tpep_dropoff_datetime']

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months

Hourly distribution:

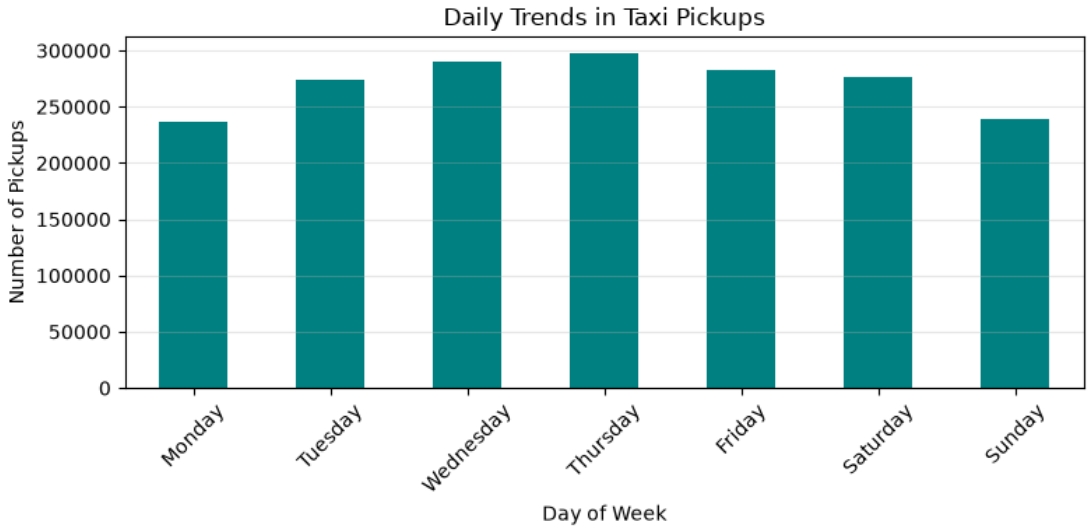


Busiest hours – 11 am to evening 10pm. Highest at 6 pm.

Low pickups: midnight 1 am to 5 am.

From 6 pm to 6 am in morning pickup count decreases. From 6 am to 6 pm pickup count increases.

Daily trend in pickup:

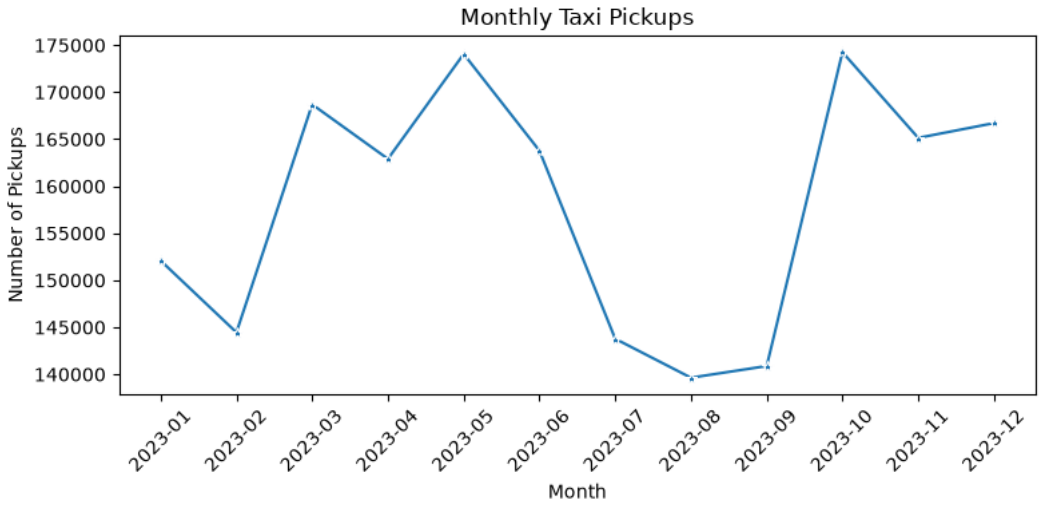


Number of pickups slightly low on Monday and Sunday. Highest on Thursday.

Monthly trend in pickup:

Pick periods where the pickups are high – March to June and Oct to Dec

Low periods where pickups are less - Jan, Feb , July to September.



3.1.3. Filter out the zero/negative values in fares, distance and tips

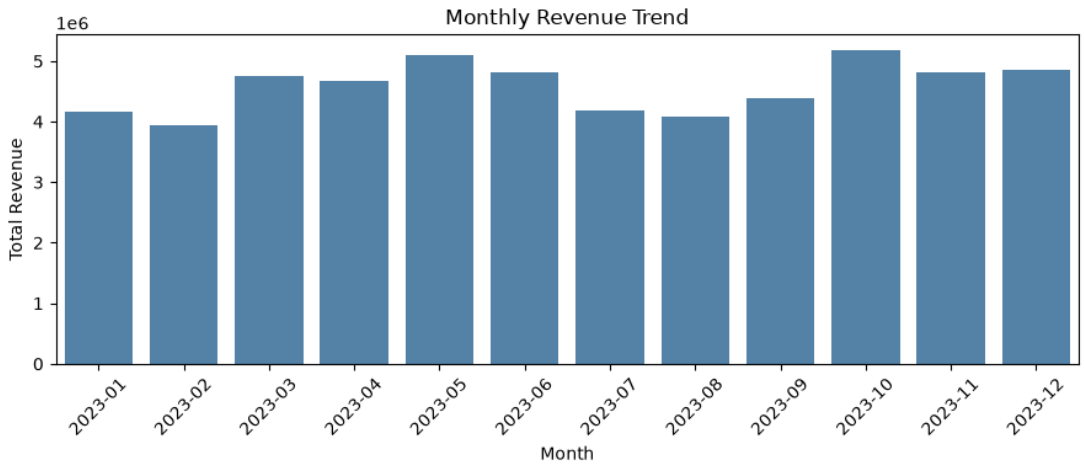
fare_amount: zeros=588, negatives=0

tip_amount: zeros=435847, negatives=0

total_amount: zeros=264, negatives=0

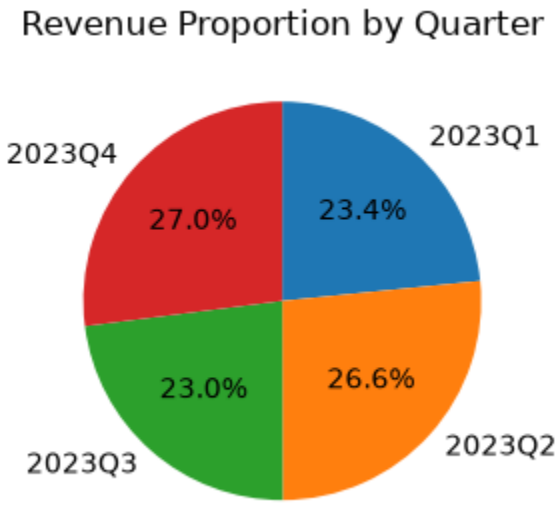
trip_distance: zeros=37625, negatives=0

3.1.4. Analyse the monthly revenue trends



Revenue is comparatively less in Jan, Feb, July, Aug, Sep than other months. But the difference is not that significant compared to difference in number of pickups for respective months.

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

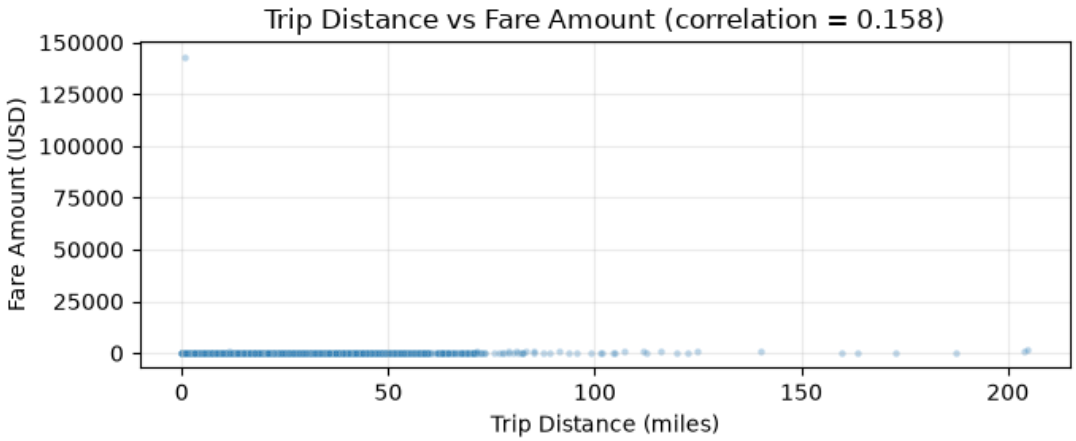


Revenue proportion is slightly high in Q2 and Q4 quarters. Q1 and Q3 quarters also perform well.

3.1.6. Analyse and visualise the relationship between distance and fare amount

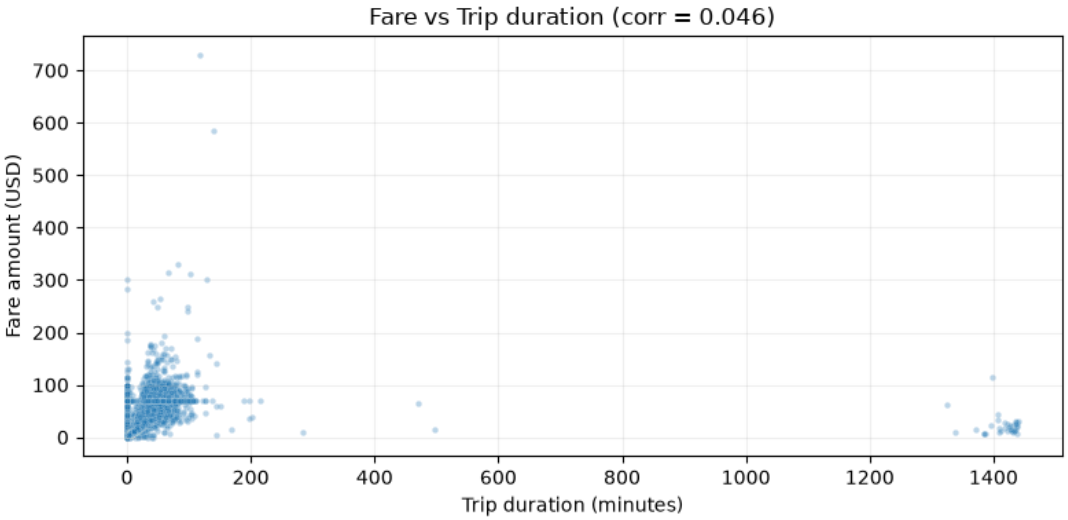
Correlation between trip_distance and fare_amount: 0.15763344357743211

trip_distance and fare_amount have week positive correlation

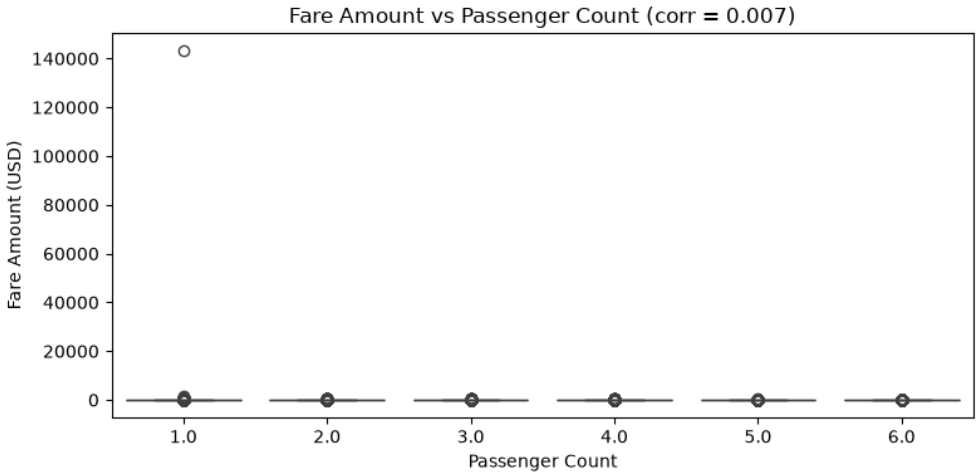


3.1.7. Analyse the relationship between fare/tips and trips/passengers

Correlation (fare_amount, trip_duration): 0.045806378258522316 which is a very week correlation, fare amount do not increase significantly with increase in trip duration.

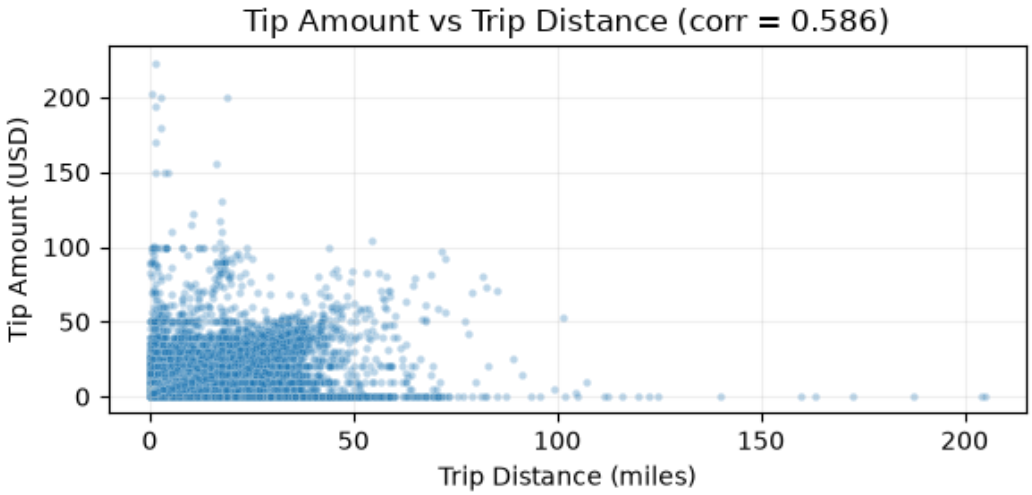


Correlation between fare_amount and passenger_count: 0.006658716115426211 – very weak correlation. More passengers doesn't lead to much increase in fare amount



Correlation between tip_amount and trip_distance: 0.5855059481686394.

For longer trip distances, tip amount received is higher. Strong positive correlation.



3.1.8. Analyse the distribution of different payment types

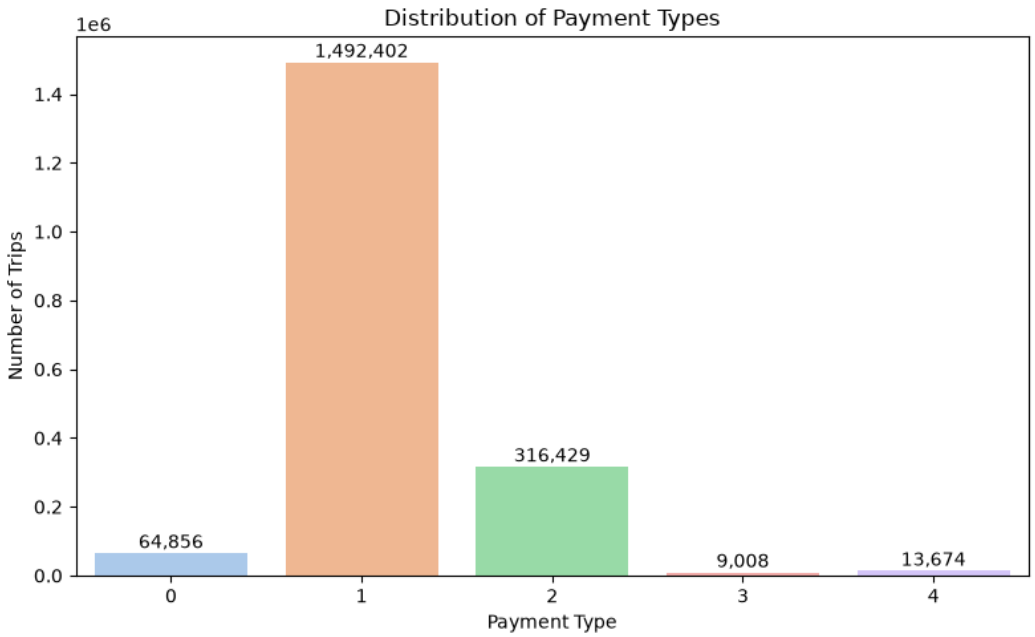
Credit card payments is used in 78% cases, followed by cash payment in 16 % cases. Undefined payment mode is used in 3 percent cases

1 - Credit card

2 - Cash

3 - No charge

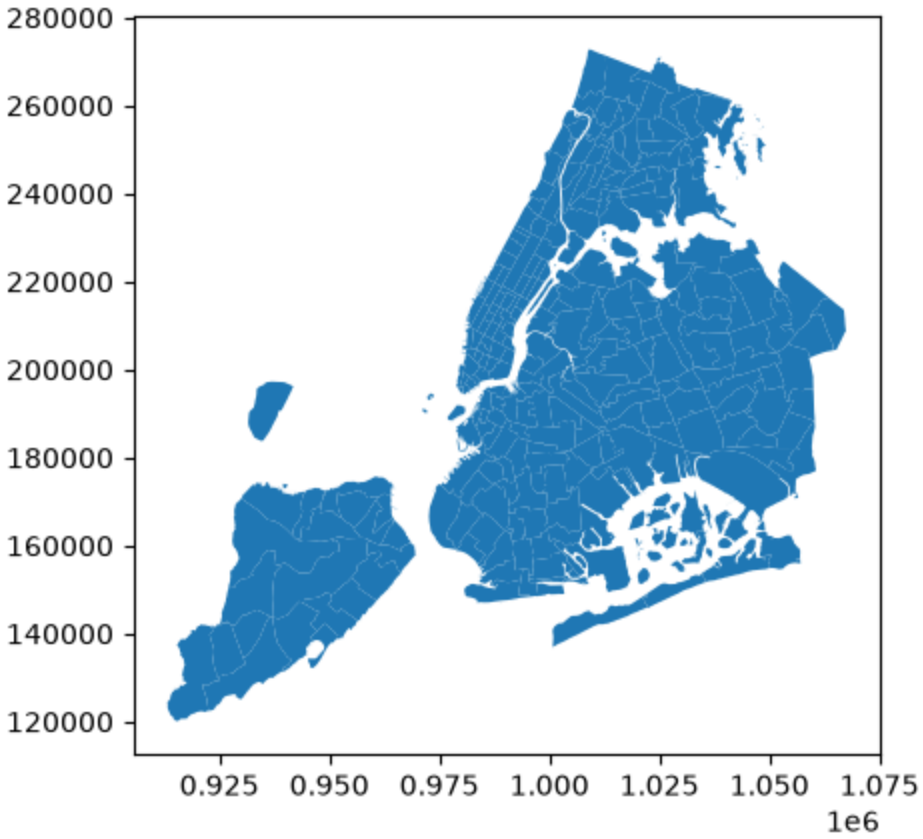
4 - Dispute



3.1.9. Load the taxi zones shapefile and display it

	OBJECTID	Shape_Leng	Shape_Area	zone	LocationID	borough	g
0	1	0.116357	0.000782	Newark Airport	1	EWR	POLYGON ((933100.918 192536.086, 933091
1	2	0.433470	0.004866	Jamaica Bay	2	Queens	MULTIPOLYGON (((1033269.244 172126.008,
2	3	0.084341	0.000314	Allerton/Pelham Gardens	3	Bronx	POLYGON ((1026308.77 256767.698, 102649
3	4	0.043567	0.000112	Alphabet City	4	Manhattan	POLYGON ((992073.467 203714.076, 992068
4	5	0.092146	0.000498	Arden Heights	5	Staten Island	POLYGON ((935843.31 144283.336, 936046.5

```
<class 'geopandas.geodataframe.GeoDataFrame'>
RangeIndex: 263 entries, 0 to 262
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   OBJECTID    263 non-null    int32
 1   Shape_Leng  263 non-null    float64
 2   Shape_Area  263 non-null    float64
 3   zone        263 non-null    str
 4   LocationID  263 non-null    int32
 5   borough     263 non-null    str
 6   geometry    263 non-null    geometry
dtypes: float64(2), geometry(1), int32(2), str(2)
memory usage: 18.5 KB
None
```



3.1.10. Merge the zone data with trips data

```
trip_df = trip_df.merge(pickup_zones, on='PULocationID', how='left')

trip_df[['PULocationID', 'PUZone', 'PUBorough']].head()
```

148] ✓ 0.5s

	PULocationID	PUZone	PUBorough
0	138	LaGuardia Airport	Queens
1	161	Midtown Center	Manhattan
2	237	Upper East Side South	Manhattan
3	143	Lincoln Square West	Manhattan
4	66	DUMBO/Vinegar Hill	Brooklyn

3.1.11. Find the number of trips for each zone/location ID

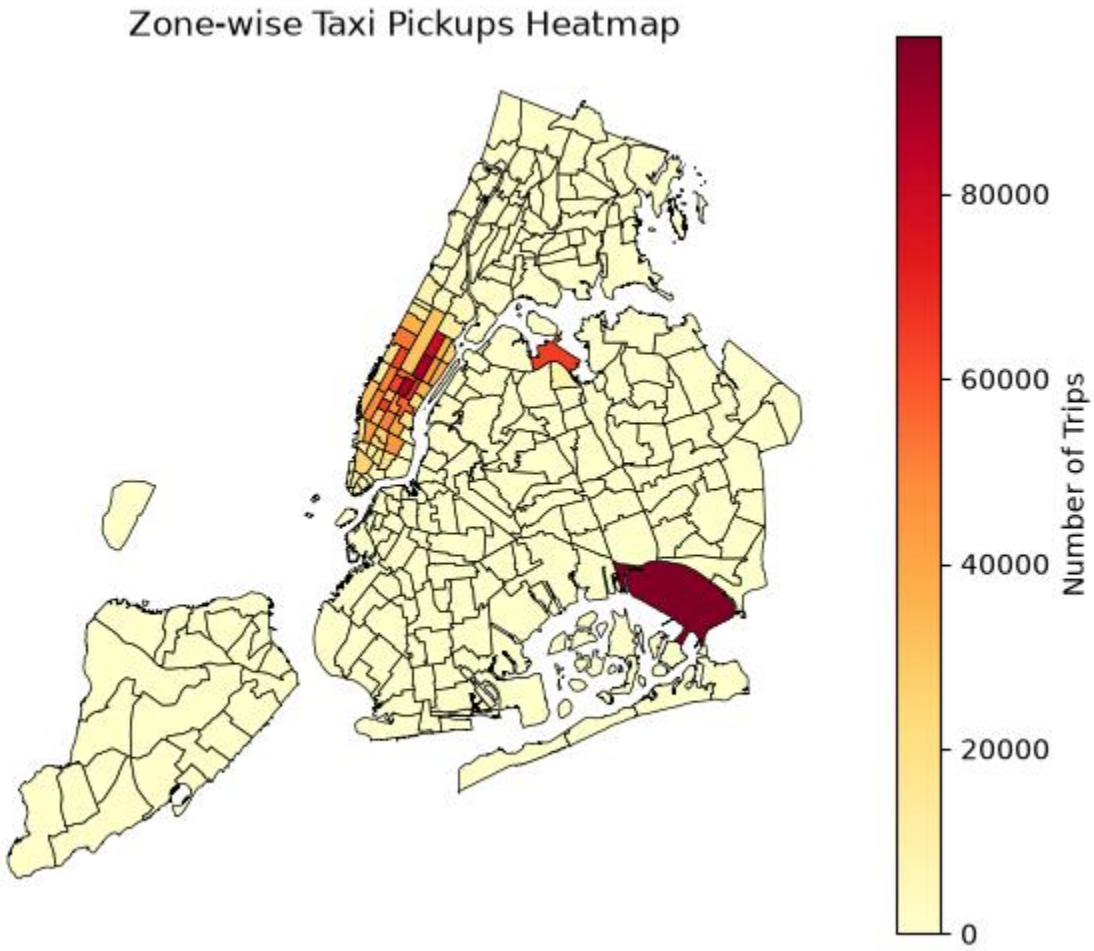
Top 5 location/zones for trips

```
PULocationID  PUZone
132           JFK Airport              97035
237           Upper East Side South    88607
161           Midtown Center           87586
236           Upper East Side North    79755
162           Midtown East             66712
Name: trip_count, dtype: int64
```

3.1.12. Add the number of trips for each zone to the zones dataframe

	LocationID	zone	borough	trip_count
0	1	Newark Airport	EWR	218.0
1	2	Jamaica Bay	Queens	2.0
2	3	Allerton/Pelham Gardens	Bronx	44.0
3	4	Alphabet City	Manhattan	2355.0
4	5	Arden Heights	Staten Island	13.0
5	6	Arrochar/Fort Wadsworth	Staten Island	25.0
6	7	Astoria	Queens	1037.0
7	8	Astoria Park	Queens	6.0
8	9	Auburndale	Queens	26.0

3.1.13. Plot a map of the zones showing number of trips



```
print(zones_sorted[['LocationID', 'zone', 'borough', 'trip_count']]
```

55] ✓ 0.0s

	LocationID	zone	borough	trip_count
131	132	JFK Airport	Queens	97035.0
236	237	Upper East Side South	Manhattan	88607.0
160	161	Midtown Center	Manhattan	87586.0
235	236	Upper East Side North	Manhattan	79755.0
161	162	Midtown East	Manhattan	66712.0
137	138	LaGuardia Airport	Queens	64441.0
185	186	Penn Station/Madison Sq West	Manhattan	64314.0
229	230	Times Sq/Theatre District	Manhattan	62669.0
141	142	Lincoln Square East	Manhattan	62662.0
169	170	Murray Hill	Manhattan	55931.0
162	163	Midtown North	Manhattan	54838.0
238	239	Upper West Side South	Manhattan	52692.0
233	234	Union Sq	Manhattan	50785.0
47	48	Clinton East	Manhattan	50323.0
67	68	East Chelsea	Manhattan	49298.0
78	79	East Village	Manhattan	45319.0
140	141	Lenox Hill West	Manhattan	44929.0
163	164	Midtown South	Manhattan	44101.0
248	249	West Village	Manhattan	41951.0
106	107	Gramercy	Manhattan	39873.0

3.1.14. Conclude with results

Based on above analysis here are the findings:

- Busiest hours – 11 am to evening 10pm. Highest at 6 pm.

Low pickups: midnight 1 am to 5 am.

From 6 pm to 6 am in morning pickup count decreases. From 6 am to 6 pm pickup count increases.

Number of pickups slightly low on Monday and Sunday. Highest on Thursday.

Pick periods where the pickups are high – March to June and Oct to Dec.

Low periods where pickups are less - Jan, Feb,July to September.

- Revenue is comparatively less in Jan, Feb, July, Aug, Sep than other months. But the difference is not that significant compared to difference in number of pickups for respective months.

- Revenue proportion is slightly high in Q2 and Q4 quarters. Q1 and Q3 quarters also perform well.

- Fare amount doesn't grow much w.r.t. trip distance or trip duration or passanger count. Though fare amount increases slightly with increare in trip distance or trip duration or passanger count. Weak positive correlation.

- Tip amount is higher when trip distance increases, both have positive co-relation.

- Busiest zones are - JFK Airport , Upper East Side South, Midtown Center, Midtown East, LaGuardia Airport, Penn Station/Madison Sq West, Times Sq/Theatre District, Lincoln Square East, Murray Hill, Midtown North

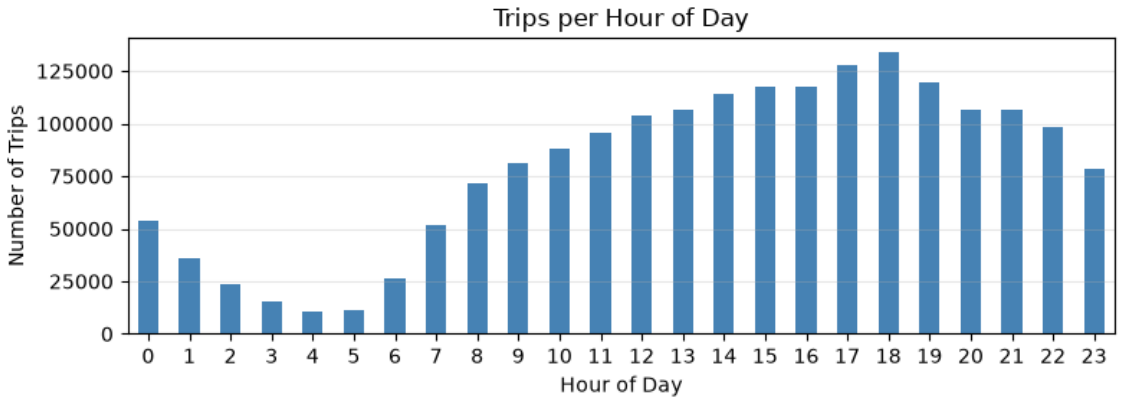
3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

Slowest routes per hour of the day –

	hour	PULocationID	PUZone	DOLocationID	DOZone	trip_count	avg_distance	avg_duration	avg_speed
0	0	234	Union Sq	113	Greenwich Village North	33	0.867879	47.832323	1.088651
1	1	148	Lower East Side	148	Lower East Side	66	0.609848	27.368687	1.336963
2	2	79	East Village	158	Meatpacking/West Village West	30	1.686667	103.230556	0.980330
3	3	148	Lower East Side	148	Lower East Side	47	0.702128	34.586879	1.218024
4	4	148	Lower East Side	79	East Village	32	0.735938	4.457292	9.906520
5	5	48	Clinton East	230	Times Sq/Theatre District	43	0.801163	4.714729	10.195659
6	6	100	Garment District	170	Murray Hill	47	0.977021	7.752482	7.561614
7	7	43	Central Park	43	Central Park	36	1.220833	45.056481	1.625737
8	8	75	East Harlem South	238	Upper West Side North	36	1.303889	47.710648	1.639746
9	9	170	Murray Hill	164	Midtown South	80	0.697625	25.140417	1.664949
10	10	90	Flatiron	186	Penn Station/Madison Sq West	60	0.687667	29.471667	1.399989
11	11	79	East Village	68	East Chelsea	36	1.970000	94.020833	1.257168
12	12	186	Penn Station/Madison Sq West	164	Midtown South	82	0.749634	31.046951	1.448711
13	13	113	Greenwich Village North	113	Greenwich Village North	45	0.461111	36.170000	0.764906
14	14	48	Clinton East	230	Times Sq/Theatre District	60	0.778333	55.054167	0.848256
15	15	100	Garment District	100	Garment District	34	0.829412	52.228922	0.952819
16	16	211	SoHo	158	Meatpacking/West Village West	33	1.386970	52.724242	1.578367
17	17	114	Greenwich Village South	113	Greenwich Village North	44	0.760227	37.319697	1.222240
18	18	114	Greenwich Village South	249	West Village	78	0.908590	43.414530	1.255694
19	19	50	Clinton West	230	Times Sq/Theatre District	33	0.940303	50.024242	1.127817
20	20	162	Midtown East	161	Midtown Center	87	0.676207	39.758046	1.020483
21	21	164	Midtown South	100	Garment District	32	0.787188	51.182812	0.922795
22	22	163	Midtown North	230	Times Sq/Theatre District	83	0.804699	42.921687	1.124884
23	23	170	Murray Hill	230	Times Sq/Theatre District	41	1.035122	43.771951	1.418884

3.2.2. Calculate the hourly number of trips and identify the busy hours



Busiest hour: 18:00

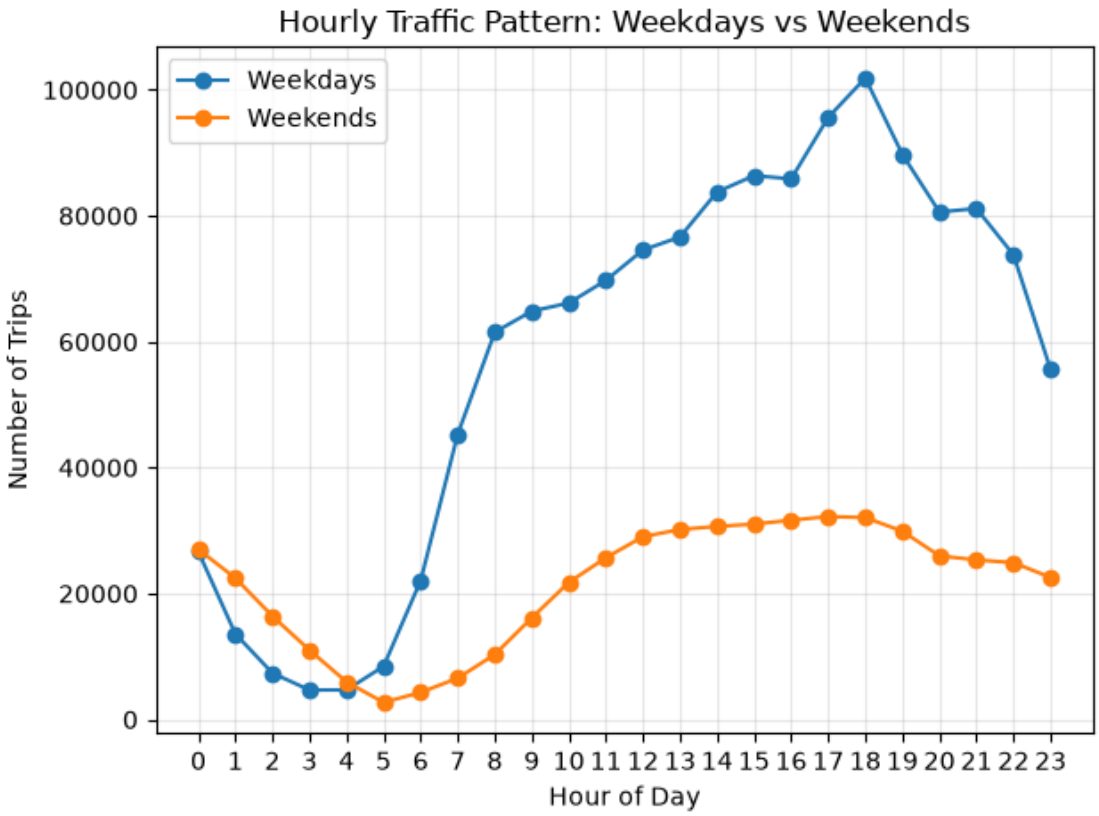
Number of trips during busiest hour: 133,931

3.2.3. Scale up the number of trips from above to find the actual number of trips

```
Sampling fraction: 0.05
Scaling factor: 20

Five busiest hours (estimated actual trips):
tpep_pickup_datetime
18    2678620
17    2557160
19    2392980
16    2349940
15    2347920
dtype: int64
```

3.2.4. Compare hourly traffic on weekdays and weekends



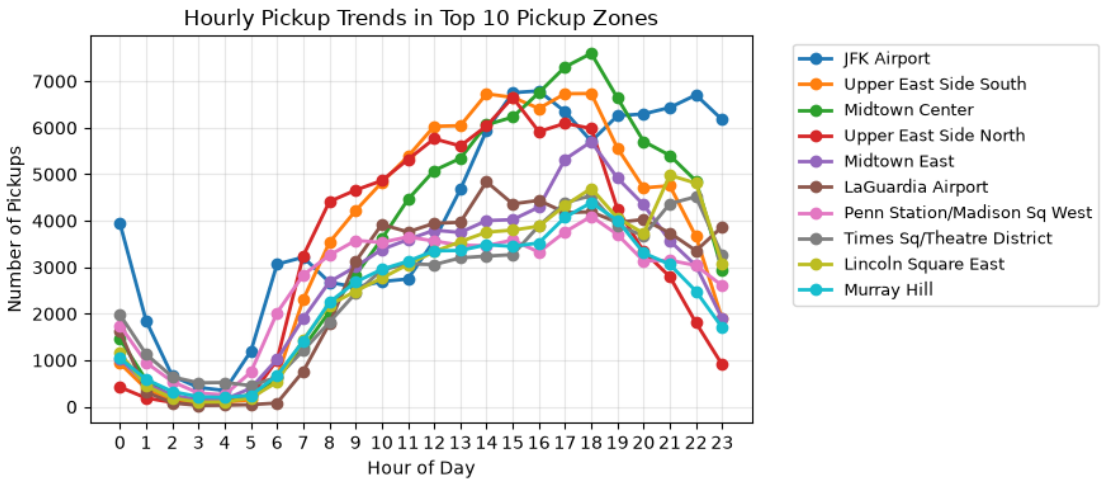
Busiest weekday hour: 18:00 (101,873 trips)

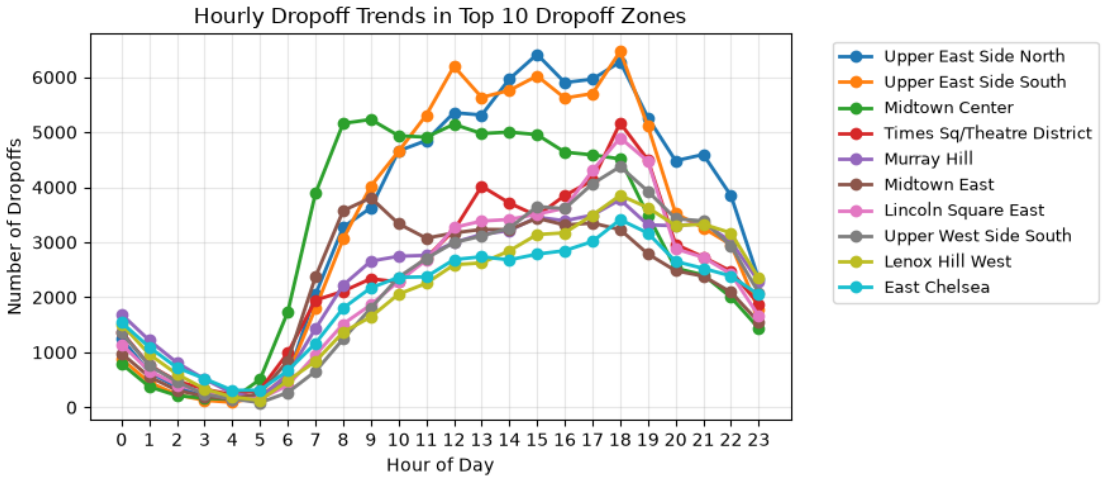
Busiest weekend hour: 17:00 (32,226 trips)

3.2.5. Identify the top 10 zones with high hourly pickups and drops

```
Top 10 Pickup Zones:
PULocationID  PUZone
132           JFK Airport                    97035
237           Upper East Side South          88607
161           Midtown Center                 87586
236           Upper East Side North          79755
162           Midtown East                   66712
138           LaGuardia Airport              64441
186           Penn Station/Madison Sq West   64314
230           Times Sq/Theatre District      62669
142           Lincoln Square East            62662
170           Murray Hill                    55931
Name: pickup_count, dtype: int64

Top 10 Dropoff Zones:
DOLocationID  PUZone
236           Upper East Side North        83442
237           Upper East Side South        79425
161           Midtown Center               73728
230           Times Sq/Theatre District    57969
170           Murray Hill                  55799
162           Midtown East                 53703
142           Lincoln Square East          53041
239           Upper West Side South        52913
141           Lenox Hill West              49858
68            East Chelsea                 47968
Name: dropoff_count, dtype: int64
```





3.2.6. Find the ratio of pickups and dropoffs in each zone

Top 10 zones by pickup/dropoff ratio:

	location_id	zone_name	pickup_drop_ratio
70	70	East Elmhurst	8.147287
128	132	JFK Airport	4.26415
134	138	LaGuardia Airport	2.620618
182	186	Penn Station/Madison Sq West	1.5569
110	114	Greenwich Village South	1.376644
42	43	Central Park	1.368644
245	249	West Village	1.332328
158	162	Midtown East	1.24224
157	161	Midtown Center	1.187961
100	100	Garment District	1.186956

Bottom 10 zones by pickup/dropoff ratio:

	location_id	zone_name	pickup_drop_ratio
172	176	Oakwood	0.0
99	99	Freshkills Park	0.0
29	30	Broad Channel	0.0
241	245	West Brighton	0.032258
0	1	Newark Airport	0.037959
111	115	Grymes Hill/Clifton	0.04
26	27	Breezy Point/Fort Tilden/Riis Beach	0.051282
253	257	Windsor Terrace	0.053952
124	128	Inwood Hill Park	0.067797
152	156	Mariners Harbor	0.068966

3.2.7. Identify the top zones with high traffic during night hours

Top 10 pickup zones (11PM to 5AM):

	PULocationID	PUZone	pickup_count
71	79	East Village	16466
116	132	JFK Airport	14584
225	249	West Village	13144
42	48	Clinton East	10834
132	148	Lower East Side	10191
100	114	Greenwich Village South	9193
207	230	Times Sq/Theatre District	8500
166	186	Penn Station/Madison Sq West	7135
147	164	Midtown South	6468
60	68	East Chelsea	6401

Top 10 dropoff zones (11PM to 5AM):

	DOLocationID	dropoff_count	DOZone
78	79	9316	East Village
46	48	7809	Clinton East
165	170	6947	Murray Hill
67	68	6549	East Chelsea
103	107	6356	Gramercy
136	141	6078	Lenox Hill West
256	263	5789	Yorkville West
242	249	5521	West Village
223	230	5367	Times Sq/Theatre District
232	239	5143	Upper West Side South

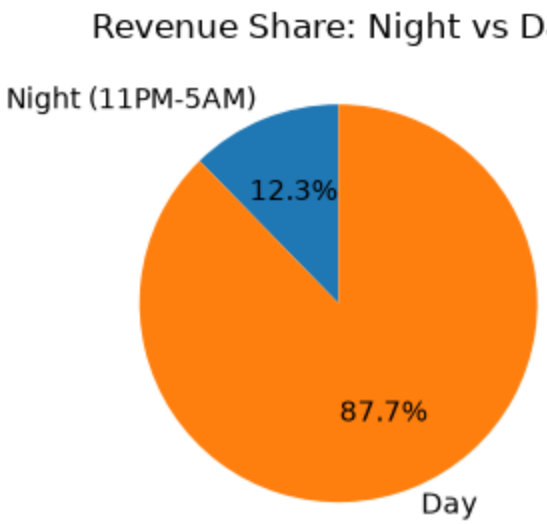
3.2.8. Find the revenue share for nighttime and daytime hours

Night revenue (11PM-5AM): $6,756,708.89

Day revenue: $48,190,030.50

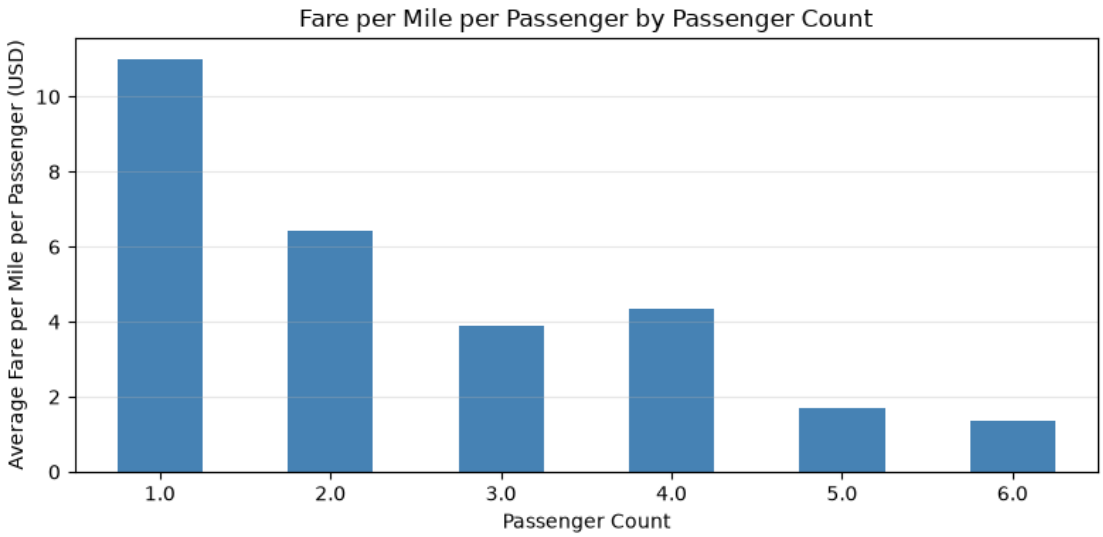
Night share: 12.30%

Day share: 87.70%



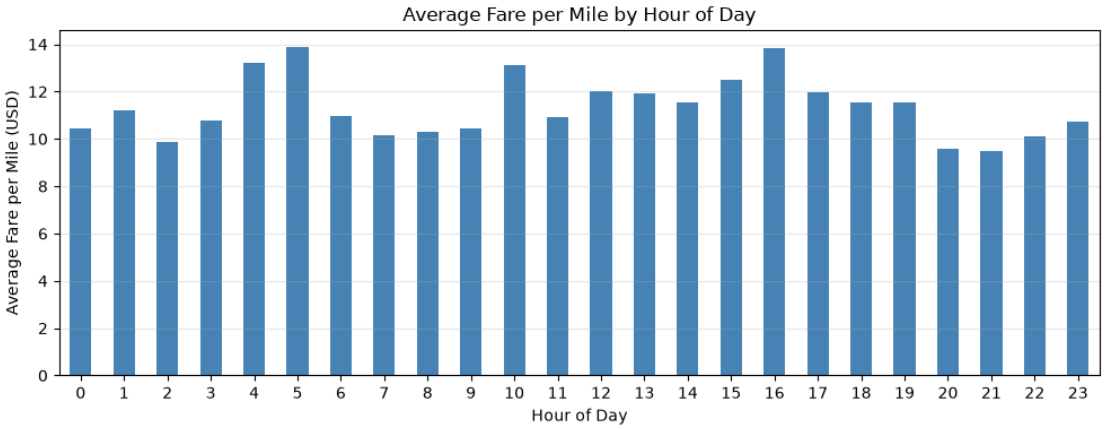
3.2.9. For the different passenger counts, find the average fare per mile per passenger

```
Average Fare per Mile per Passenger by Passenger Count:
                 avg_fare_per_mile_per_passenger
passenger_count
1.0                                    11.003255
2.0                                     6.434912
3.0                                     3.907743
4.0                                     4.360870
5.0                                     1.709582
6.0                                     1.350662
```

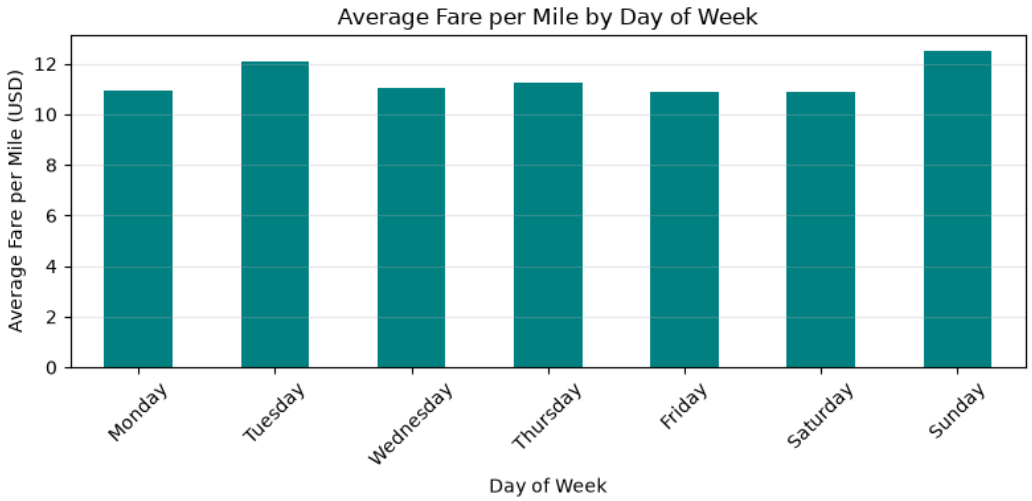


3.2.10. Find the average fare per mile by hours of the day and by days of the week

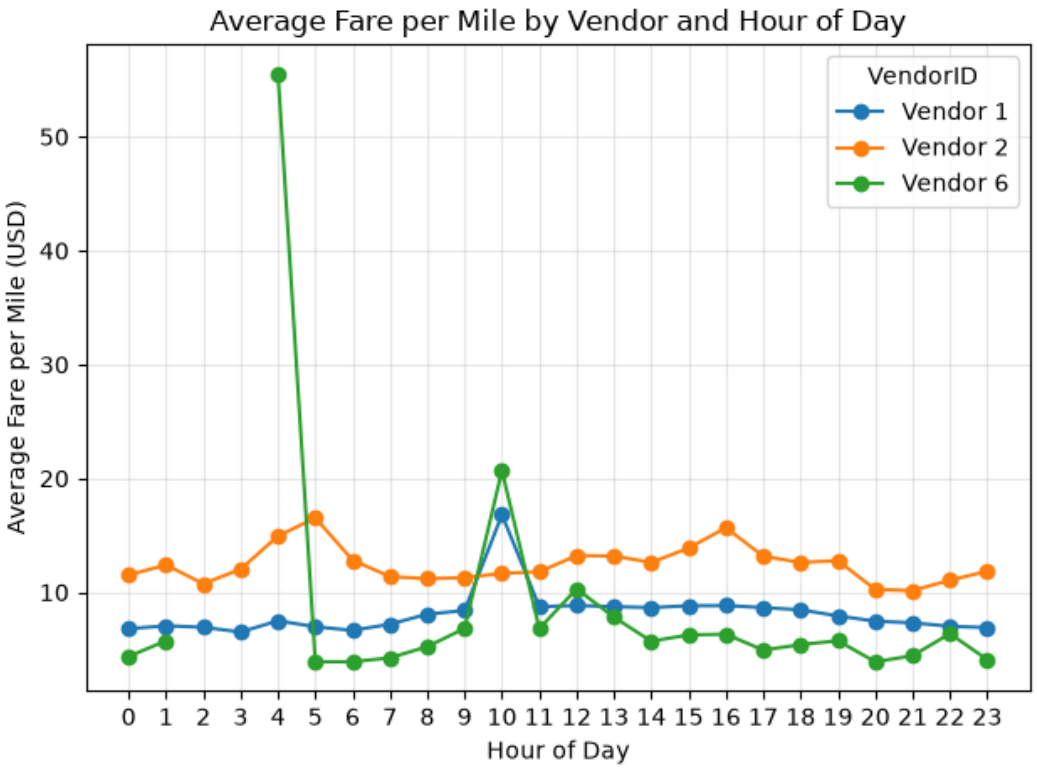
```
Average Fare per Mile by Hour of Day:
hour
0     10.466255
1     11.202882
2      9.871727
3     10.799226
4     13.228243
5     13.890256
6     10.984566
7     10.162749
8     10.305552
9     10.463414
10    13.102917
11    10.937973
12    12.009172
13    11.937688
14    11.523513
15    12.495522
16    13.813875
17    11.964355
18    11.545668
19    11.550656
20     9.559832
21     9.484459
22    10.124651
23    10.709787
Name: fare_per_mile, dtype: float64
```



```
Average Fare per Mile by Day of Week:
day_of_week
Monday       10.925552
Tuesday      12.081706
Wednesday    11.039181
Thursday     11.239192
Friday       10.903320
Saturday     10.871707
Sunday       12.494536
Name: fare_per_mile, dtype: float64
```

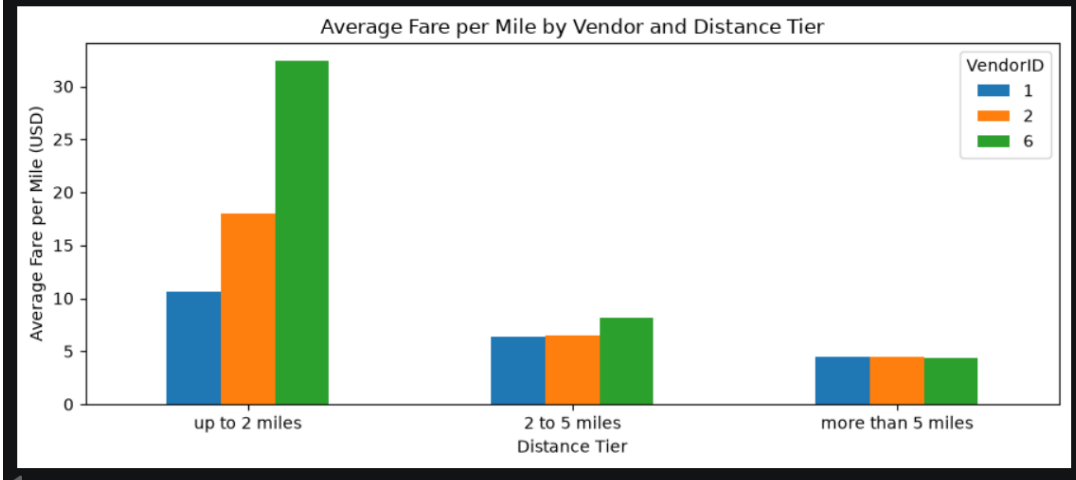


3.2.11. Analyse the average fare per mile for the different vendors

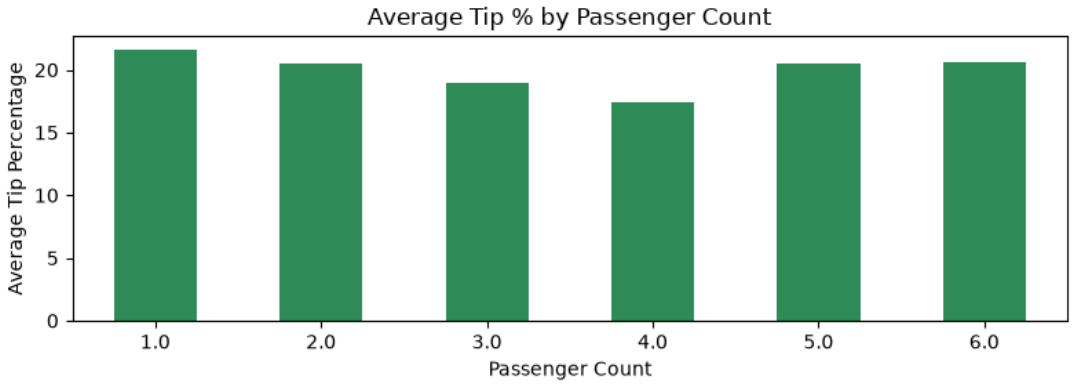


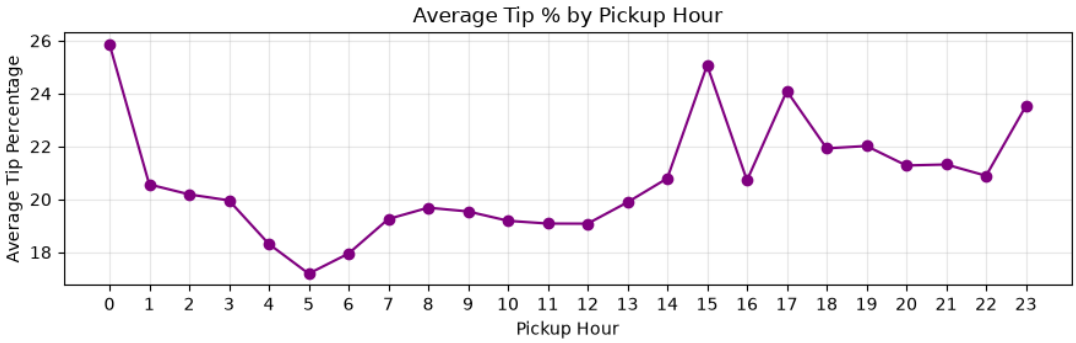
3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

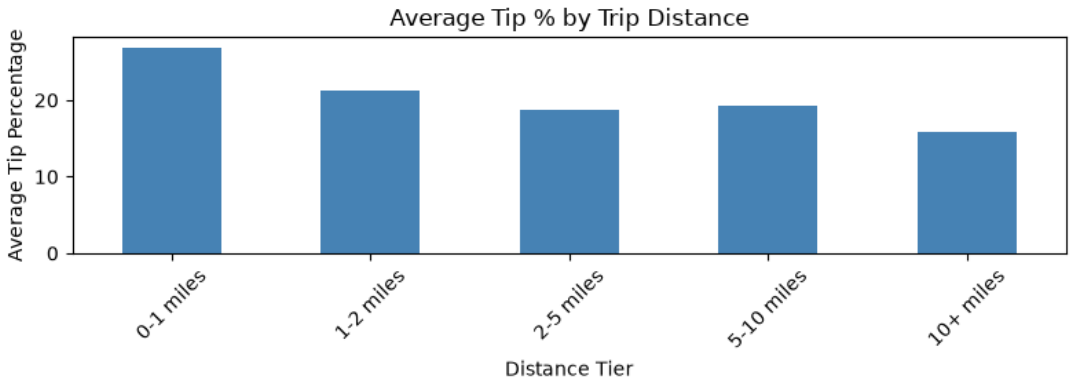
VendorID	1	2	6
distance_tier			
up to 2 miles	10.635032	18.005383	32.422471
2 to 5 miles	6.380893	6.537991	8.107119
more than 5 miles	4.423563	4.490261	4.382026



3.2.13. Analyse the tip percentages


Average Tip % by Passenger Count
Average Tip Percentage
0
5
10
15
20
1.0
2.0
3.0
4.0
5.0
6.0
Passenger Count


Average Tip % by Pickup Hour
Average Tip Percentage
18
20
22
24
26
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Pickup Hour


Average Tip % by Trip Distance
Average Tip Percentage
0
10
20
0-1 miles
1-2 miles
2-5 miles
5-10 miles
10+ miles
Distance Tier

```
Average tip % by trip distance tier: distance_tier
0-1 miles     26.872089
1-2 miles     21.191221
2-5 miles     18.639749
5-10 miles    19.266736
10+ miles     15.813881
Name: tip_pct, dtype: float64

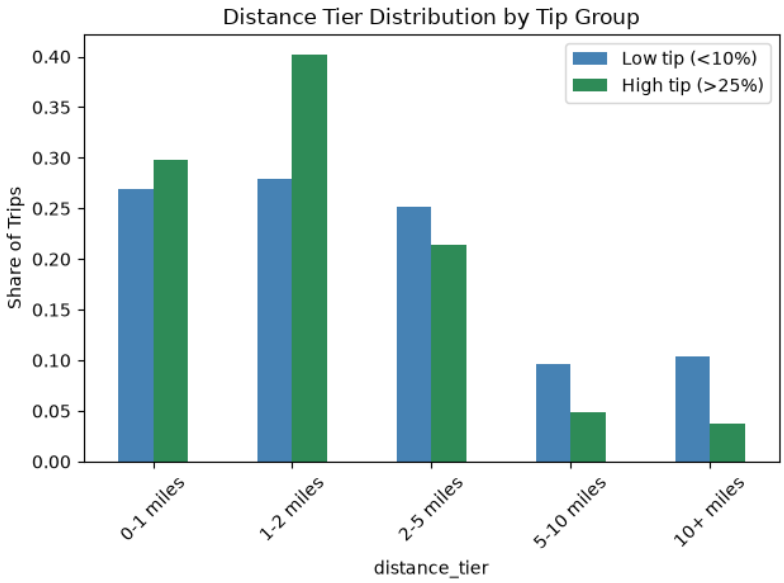
Average tip % by passenger count: passenger_count
1.0    21.592003
2.0    20.455295
3.0    18.967309
4.0    17.379818
5.0    20.522174
6.0    20.581536
Name: tip_pct, dtype: float64

Average tip % by pickup hour: pickup_hour
0     25.872895
1     20.562824
2     20.186266
3     19.958252
4     18.311512
5     17.187506
6     17.947651
7     19.255079
8     19.682646
9     19.540640
10    19.187722
```

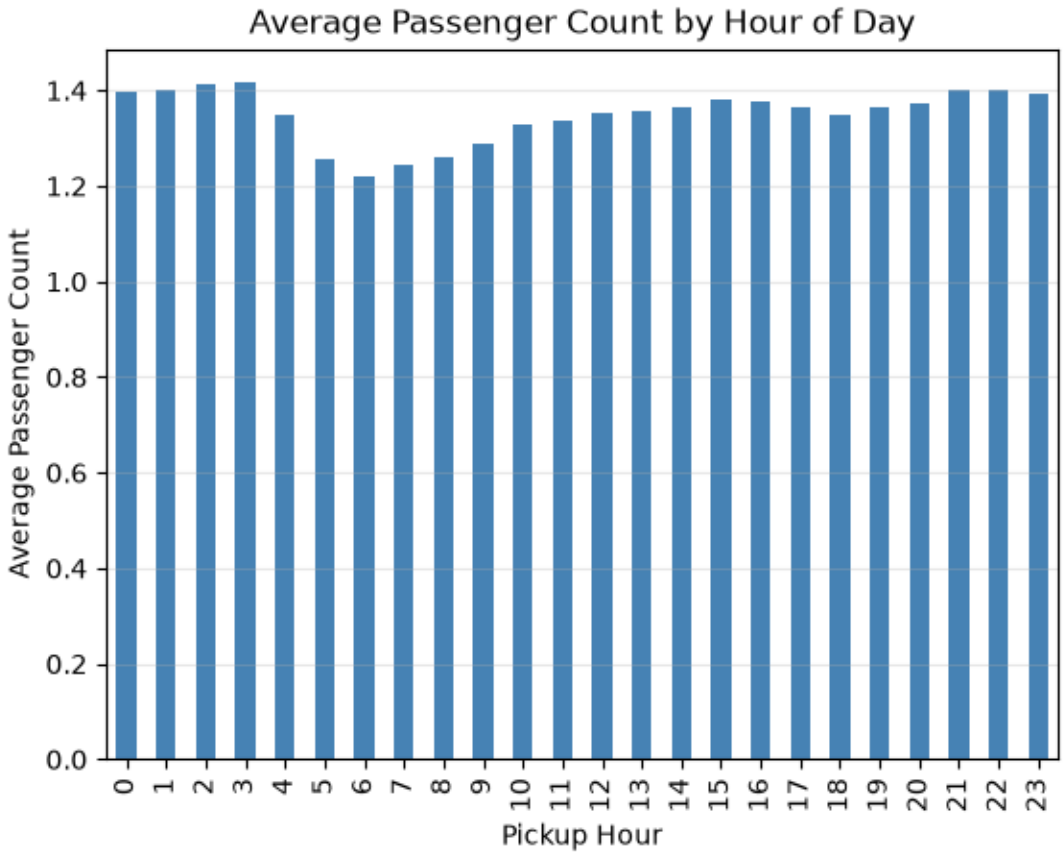
```
Low-tip trips (<10%): 505951
High-tip trips (>25%): 815534

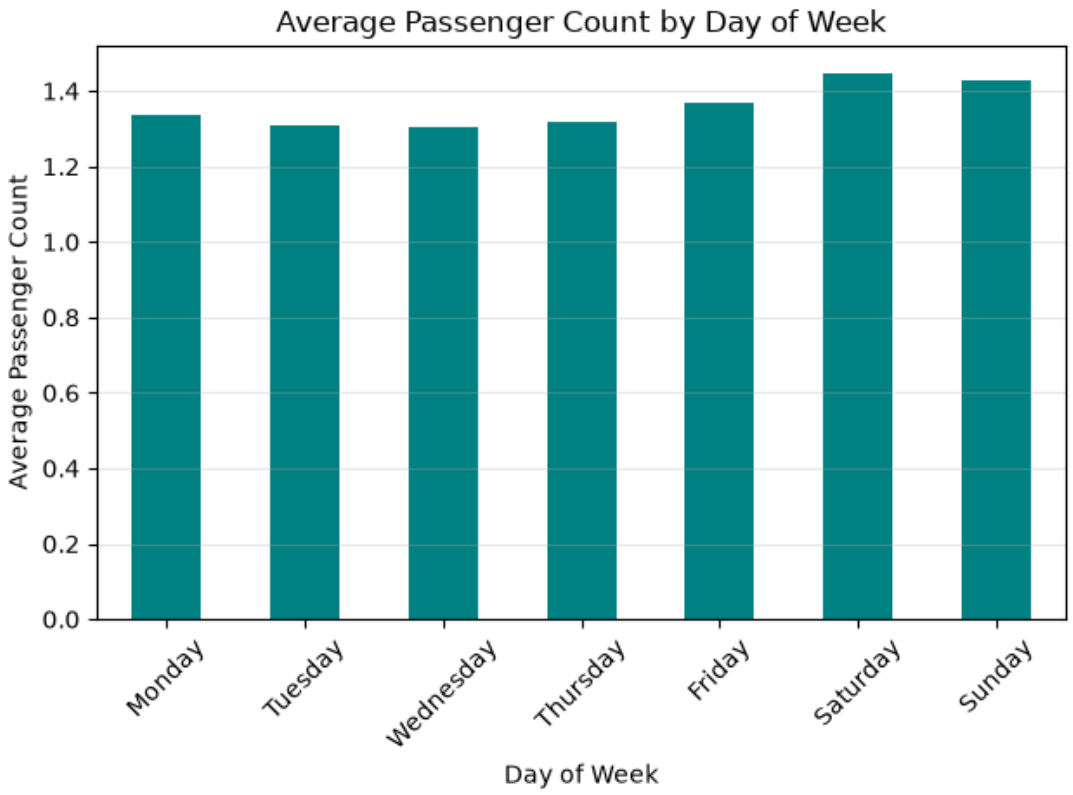
Summary comparison:
                    Low tip (<10%) High tip (>25%)
count                       505951          815534
avg_fare_amount              22.09           14.48
avg_trip_distance             3.73            2.31
avg_passenger_count           1.41            1.36
mode_pickup_hour                15              18
mode_payment_type                2               1
mode_vendor                      2               2
mode_distance_tier       1-2 miles       1-2 miles

Distance tier share:
               Low tip (<10%)  High tip (>25%)
distance_tier
0-1 miles            0.268593         0.297703
1-2 miles            0.279640         0.401445
2-5 miles            0.251297         0.213659
5-10 miles           0.096256         0.049227
10+ miles            0.104214         0.037967
```

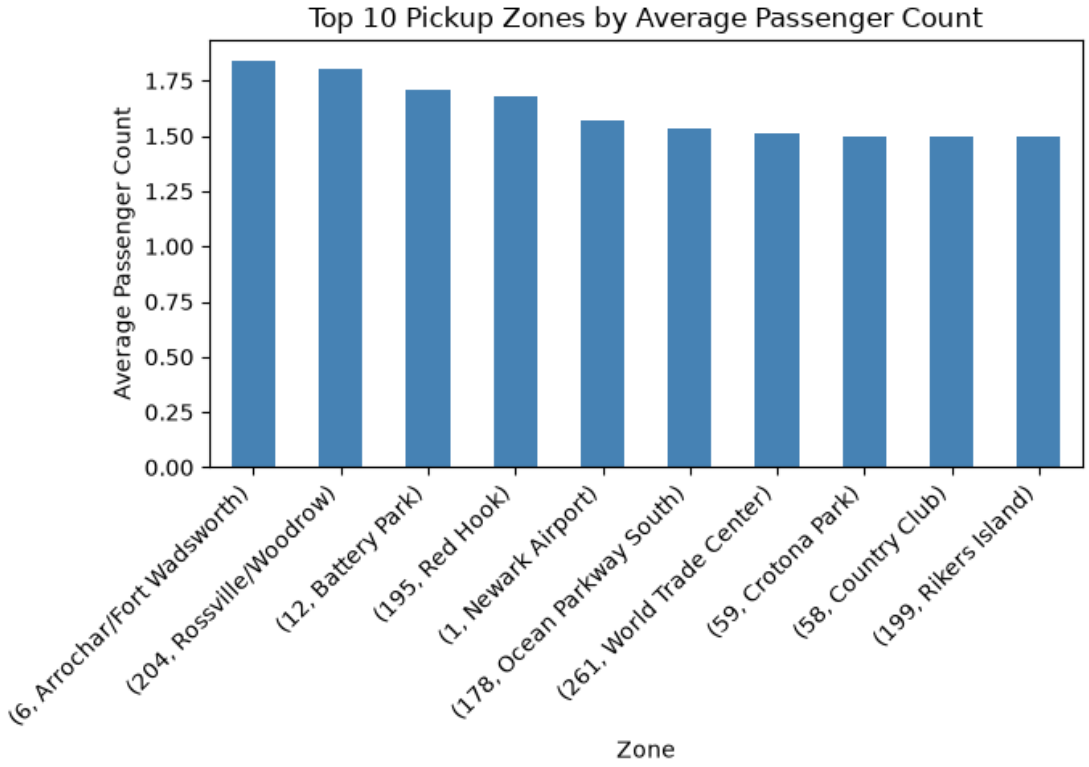


3.2.14. Analyse the trends in passenger count




Average Passenger Count by Day of Week
Average Passenger Count
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day of Week

3.2.15. Analyse the variation of passenger counts across zones



Top 10 zones by average passenger count:

PULocationID	PUZone	avg_passengers
6	Arrochar/Fort Wadsworth	1.840000
204	Rossville/Woodrow	1.800000
12	Battery Park	1.708680
195	Red Hook	1.682609
1	Newark Airport	1.568807
178	Ocean Parkway South	1.533333
261	World Trade Center	1.512178
59	Crotona Park	1.500000
58	Country Club	1.500000
199	Rikers Island	1.500000

	OBJECTID	Shape_Leng	Shape_Area	zone	LocationID	borough	geometry	trip_count	avg_passenger_count
0	1	0.116357	0.000782	Newark Airport	1	EWR	POLYGON ((933100.918 192536.086, 933091.011 19...	218.0	1.568807
1	2	0.433470	0.004866	Jamaica Bay	2	Queens	MULTIPOLYGON (((1033269.244 172126.008, 103343...	2.0	1.000000
2	3	0.084341	0.000314	Allerton/Pelham Gardens	3	Bronx	POLYGON ((1026308.77 256767.698, 1026495.593 2...	44.0	1.022727
3	4	0.043567	0.000112	Alphabet City	4	Manhattan	POLYGON ((992073.467 203714.076, 992068.667 20...	2355.0	1.323567
4	5	0.092146	0.000498	Arden Heights	5	Staten Island	POLYGON ((935843.31 144283.336, 936046.565 144...	13.0	1.000000

```
                      trips_with_charge  pct_trips_with_charge  total_amount
charge
extra                           1139587              60.093104  3.011519e+06
mta_tax                         1878661              99.066215  9.393502e+05
tolls_amount                     153832               8.111923  1.131465e+06
improvement_surcharge           1895521              99.955283  1.894624e+06
congestion_surcharge            1755486              92.570908  4.376240e+06
airport_fee                      225811              11.907545  2.710245e+05

Trips with any surcharge/extra charge: 1895848

Trips % with any surcharge (99.97252644395684)
```

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

```
Extra charge type - extra: 1139587 trips (60.09% of all trips)
Top pickup zones:
PUZone
LaGuardia Airport    63576
Name: count, dtype: int64
Top dropoff zones:
DOZone
Corona    2
Name: count, dtype: int64
Top pickup hours:
tpep_pickup_datetime
18    104817
Name: count, dtype: int64
Top dropoff hours:
tpep_dropoff_datetime
18    108416
Name: count, dtype: int64
Extra charge type - mta_tax: 1878661 trips (99.07% of all trips)
Top pickup zones:
PUZone
JFK Airport    94964
Name: count, dtype: int64
Top dropoff zones:
DOZone
Governor's Island/Ellis Island/Liberty Island    3
Name: count, dtype: int64
Top pickup hours:
tpep_pickup_datetime
18    132936
Name: count, dtype: int64
```

```
Top dropoff hours:
tpep_dropoff_datetime
18    137091
Name: count, dtype: int64
Extra charge type - tolls_amount: 153832 trips (8.11% of all trips)
Top pickup zones:
PUZone
JFK Airport    47061
Name: count, dtype: int64
Top dropoff zones:
DOZone
Bayside    1
Name: count, dtype: int64
Top pickup hours:
tpep_pickup_datetime
14    11874
Name: count, dtype: int64
Top dropoff hours:
tpep_dropoff_datetime
16    11555
Name: count, dtype: int64
Extra charge type - improvement_surcharge: 1895521 trips (99.96% of all trips)
Top pickup zones:
PUZone
JFK Airport    96998
Name: count, dtype: int64
Top dropoff zones:
DOZone
Governor's Island/Ellis Island/Liberty Island    3
Name: count, dtype: int64
```

```
Top pickup hours:
tpep_pickup_datetime
18    133879
Name: count, dtype: int64
Top dropoff hours:
tpep_dropoff_datetime
18    138190
Name: count, dtype: int64
Extra charge type - congestion_surcharge: 1755486 trips (92.57% of all trips)
Top pickup zones:
PUZone
Upper East Side South    88377
Name: count, dtype: int64
Top dropoff zones:
DOZone
Governor's Island/Ellis Island/Liberty Island    3
Name: count, dtype: int64
Top pickup hours:
tpep_pickup_datetime
18    125991
Name: count, dtype: int64
Top dropoff hours:
tpep_dropoff_datetime
18    129721
Name: count, dtype: int64
Extra charge type - airport_fee: 225811 trips (11.91% of all trips)
Top pickup zones:
PUZone
JFK Airport    90618
Name: count, dtype: int64
```

```
Top dropoff zones:
DOZone
Newark Airport    1
Name: count, dtype: int64
Top pickup hours:
tpep_pickup_datetime
17    14845
Name: count, dtype: int64
Top dropoff hours:
tpep_dropoff_datetime
18    15904
Name: count, dtype: int64
```

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

- Peak hours
  - Deploy more vehicles during peak demand windows (especially 4PM-7PM and 10AM-3PM) and reduce fleet size during quiet night hours (11PM-5AM).
- Driver availability
  - Maintain higher driver availability Mon-Fri 7AM-9PM. Over weekend flexible scheduling and maintenance windows as demand is low.
- Vehicle deploymnet
  - Position supply near top pickup zones like JFK Airport, LaGuardia, Midtown, Upper East Side, Penn Station, and night hotspots such as East Village and Murray Hill.
- High pickup/low dropoff Ratios indicate supply concentration opportunities - East Elmhurst , JFK Airport, LaGuardia Airport. Implement surge pricing during peak hours
- Night dispatching
  - Prioritize night dispatching in high drop off zone - East Village, Clinton East, Murray Hill (top 3 dropoff zones). Deploy ride-pooling options to reduce vehicle count while maintaining service
- Rerouting
  - Monitor slow routes by hour and reroute congested trips to improve turnaround and reduce idle time.Suggest alternative routes during 4PM-7PM and redirect drivers via less congested corridors
- Car types
  - Use smaller cars for short-distance trips(0-2 miles - 51% of total trips) and reserve larger or premium vehicles for airport and longer-distance routes.
- Surge pricing
  - To optimize revenue implement surge pricing. +20-25% (4PM-7PM), +10-15%(7AM-10AM, 10PM-11PM), other times standard rates, Discount at night hours.
- Promotional prising
  - During quarters Q1 and Q3 to increase demand
- Balance supply using pickup/dropoff ratio insights to prevent oversupply in some zones and shortages in others.
- Standardize pricing across vendors or justify premium for superior service metrics
- Extra Charges: Only 60.09% of trips - underutilized revenue opportunity. Increase extra charge frequency during peak hours (4PM-7PM) for rush period premium

- Cash payments (payment type 2) associated with 67% of low-tip trips, Incentivize card payments (5% lower fare or bonus points) to improve tip capture

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

- Peak-hour staging
  - Pre-position drivers 20–30 minutes before peak (16:00–19:00) in top pickup zones (Midtown Center, Midtown East, Upper East Side South, Penn Station, Times Sq) to shorten wait times and increase throughput.

- Airport handling
  - Maintain dedicated airport fleets at JFK and LaGuardia with separate queueing; rotate drivers to/from nearby high pickup zones to avoid deadhead miles.

- Use pickup/dropoff ratio to rebalance
  - Move idle vehicles from high-drop / low-pick zones to high-pick / low-drop zones (e.g., shift capacity toward East Elmhurst, JFK, LaGuardia during peak departures).

- Night coverage (23:00–05:00)
  - Concentrate night drivers in high night-time pickup/dropoff zones (East Village, Clinton East, Murray Hill, East Chelsea, TriBeCa) and keep a small roaming fleet for airport late arrivals.

- Micro-hubs and staging clusters
  - Create micro-hubs near clusters of high-demand zones (e.g., Midtown cluster, Upper East/West) for quick redeployment and driver rest/vehicle swap.

- Fleet mix by zone & distance tiers
  - Assign small, fuel-efficient cars to short-trip dense zones (0–2 miles); sedans for 2–5 mile corridors; larger/premium vehicles for airport and long-haul trips (>5 miles).

- Dynamic repositioning & demand prediction
  - Use short-term demand forecasts (1–2 hours) to reallocate drivers; trigger automatic repositioning when predicted supply-demand imbalance exceeds threshold.

- Incentives & driver routing

- Offer relocation bonuses to drivers who reposition to undersupplied high-pickup zones before peaks; reward drivers taking pooled/shared rides during off-peak to improve utilization.

- Time & day-specific policies
  - Increase supply in March–June and Oct–Dec during historically higher demand; schedule maintenance and training in Jan–Feb and Jul–Sep lower-demand windows.

- Congestion-aware routing
  - Route drivers around identified slow corridors during peak hours; prioritize routes with better throughput for airport-bound trips to reduce delay penalties.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

- Use distance-tier pricing:
  - Short trips (up to 2 miles) account for the largest share of rides, so keep a slightly higher per-mile floor for that tier to cover fixed pickup costs, while offering more competitive per-mile pricing for 2–5 mile and >5 mile trips.
- Apply surge only at true peaks:
  - Data shows highest demand around 2PM–7PM and strong weekday morning/evening peaks, so set +20–25% for late afternoon rush and +10–15% for 7–10 AM / 7–9 PM, while keeping midday and off-peak rates stable.
- Preserve market share at night:
  - Night revenue is lower than daytime, so avoid aggressive night surges; instead offer modest incentives or loyalty discounts for 23:00–05:00 to improve utilization in top night zones such as East Village, Clinton East, and Murray Hill.
- Vendor pricing parity:
  - Average fare-per-mile varies by vendor and distance tier, so standardize base fare structure across VendorID 1,2,6 and reserve premium pricing only for verified service advantages or faster fulfillment.
- Leverage surcharge visibility:
  - Congestion and airport fees are already common, so ensure these are transparently bundled and not hidden; consider dynamic airport/additional charge policies only for actual airport or congestion-heavy trips to avoid customer pushback.
- Encourage digital payments:

- Low-tip trips are more frequent with cash payments, so offer small fare credits or rewards for card/digital transactions to improve overall revenue capture and reduce tip leakage.
- Use quarter and zone insights:
  - Since revenue is stronger in Q2/Q4 and top pickup zones are JFK, LaGuardia, Midtown, and Upper East Side, align price adjustments with seasonality and maintain competitive rates in high-density zones to retain volume.